

Entropy-Based Causal Analysis of Precursors to Runway Excursions

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1 Abstract

Runway excursions are among the most frequent and safety-critical events in aviation, often resulting in aircraft damage, operational disruptions, and increased risk during landing and takeoff phases. Understanding the conditions that precede these events is important for improving situational awareness and supporting safety-focused decision-making in aviation operations.

This study uses a dataset of 428 runway excursion events compiled from National Transportation Safety Board (NTSB) accident records, combined with corresponding meteorological observations from METAR reports. Each observation includes environmental and operational context. The analysis applies conditional mutual information and causation entropy to identify variables that provide unique informational value about runway excursion severity. These methods produce a ranked set of precursors and inferred directed dependency graph representing conditional mutual information relationships among variables.

Results show that wind direction and altimeter settings are the most consistent and stable precursors, followed by operational variables such as FAR classification and flight purpose. Stability analysis indicates that these variables remain robust across resampling and modeling variations, while weaker predictors display less consistency. These findings suggest that runway excursion risk is driven by a combination of environmental and operational conditions rather than a single dominant factor.

Practically, this framework can support aviation safety monitoring by highlighting conditions associated with elevated risk, informing training focus areas, and contributing to decision-support systems that integrate both environmental and operational context.

2 Summary

This study investigated the informational structure underlying runway excursion severity using an information-theoretic feature selection and causal discovery framework. The primary objective was to identify variables that retain unique and stable informational value about runway excursion outcomes while accounting for dependencies among predictors.

The analysis began by constructing a binary target variable representing damage severity and evaluating candidate precursors using mutual information, conditional mutual information, and causation entropy. A greedy forward selection procedure was applied to iteratively build a conditioning set, ensuring that each newly selected variable contributed incremental information beyond previously selected features. Statistical dependence was assessed using permutation testing, while bootstrap resampling was used to evaluate the stability of selected features across repeated samples.

Across all selection methods, a consistent subset of variables emerged as the most informative. Wind direction (`drct`) and altimeter setting (`alti`) were identified as the strongest and most stable contributors to runway excursion severity. Operational variables, particularly FAR classification (`far`) and purpose of flight (`purposeofflight`), also demonstrated consistent informational value, suggesting that both environmental and operational factors contribute meaningfully to the observed outcome structure. Rather than relying on a single dominant predictor, the results indicate a distributed dependency pattern where multiple moderate-strength variables jointly contribute to runway excursion severity.

2.1 Dataset and Problem Context

The dataset used in this analysis consists of 428 runway excursion events compiled from publicly available NTSB accident records, supplemented with corresponding meteorological observations from METAR reports. Each event includes a combination of environmental variables (such as wind speed, wind direction, temperature, dew point, and pressure) and operational variables (such as flight purpose, aircraft classification, and regulatory category).

Each observation is aligned to a binary outcome variable:

- `damage_binary` = 1: substantial or destroyed aircraft damage
- `damage_binary` = 0: minor, no damage, or not applicable outcomes

This framing allows the analysis to focus specifically on identifying conditions associated with higher-severity runway excursions, rather than treating all excursions as equivalent events.

2.2 Definition of "Precursor" in This Study

In this analysis, a *precursor* is defined in a strictly informational sense. A variable is considered a precursor if it provides measurable, non-redundant information about runway excursion severity after conditioning on other observed variables [24]. Formally, a precursor is a feature that maintains statistically significant conditional dependence with the target variable under mutual information and causation entropy analysis. This means that even after accounting for other variables already included in the model, the feature still improves the explanation of the outcome.

Importantly, this definition does not imply physical causation or deterministic influence. Instead, it reflects the idea that a precursor is a variable that consistently carries *additional predictive structure* about the outcome beyond what is already captured by the dataset. In this sense, precursors are best interpreted as *informational signals* rather than direct causal drivers of runway excursion events.

2.3 Operational Meaning of "Causal"

The term *causal* in this analysis is used operationally to describe directed conditional dependencies derived from data, not philosophical or mechanistic causation. Specifically, causation entropy is utilized to construct a directed structure in which a variable is considered to have a causal influence on the target if it contributes incremental information about that target after conditioning on previously selected variables.

Thus, in this framework:

- "Causal edge" = statistically directed conditional dependence
- "Causal influence" = additional predictive information under conditioning
- "Causal graph" = structured representation of dependency relationships

These relationships are estimated from observational data, meaning they may reflect confounding effects, indirect pathways, or shared underlying drivers rather than direct physical causation. As a result, the causal graph should be interpreted as a *dependency structure supported by information-theoretic evidence*, not as a definitive model of real-world causal mechanisms.

2.4 Methodological Framework

The analysis is built around a sequential feature selection process guided by information theory. Mutual information (MI) measures overall dependence between each feature and runway excursion severity, while Conditional Mutual Information (CMI) captures the additional contribution of each feature after accounting for variables already selected. Causation Entropy (CSE) extends this by quantifying directed incremental information in a sequential framework, allowing the method to isolate variables that provide unique informational value beyond redundancy in the dataset.

At each step, the feature contributing the most additional information about the target is selected, and the process continues until no statistically significant improvement is observed or a stopping criterion is reached. This results in a compact set of variables that jointly explain the structure of the outcome without duplicating information across predictors.

Robustness is ensured through permutation testing, which evaluates whether observed dependencies exceed what would be expected under independence [18], and bootstrap resampling, which measures how consistently features are selected across repeated samples [26]. Together, these methods ensure that identified precursors are both statistically meaningful and stable under variation.

2.5 Evaluation of Success

Success in this study is evaluated based on consistency, stability, and interpretability of the inferred informational structure. The analysis is considered successful if key variables remain statistically significant under permutation testing, selected features demonstrate high stability across bootstrap resampling, the inferred dependency structure remains consistent under reasonable methodological variations, and the resulting relationships align with plausible aviation safety mechanisms without relying on spurious correlations. Overall, success is not defined by producing a single predictive model, but by identifying a compact and robust set of variables that consistently carry informational relevance about runway excursion severity. The final output is therefore interpreted as a structured representation of dependency relationships that can support further investigation, hypothesis generation, and potential applications in aviation safety contexts.

3 Problem Statement and Research Questions

Runway excursions (REs) are defined as aviation safety events in which an aircraft departs the runway surface during takeoff or landing, either by veering off the side of the runway or overrunning its end [5, 4]. Veer-offs typically occur when directional control is lost on landing or takeoff, whereas overruns occur when the aircraft fails to stop within the available runway length [5]. These events represent the most prevalent type of aviation accident over the last ten years, accounting for a significant portion of aircraft accidents during critical phases of flight [6]. Contributing factors include environmental conditions (e.g., weather, runway surface contamination), operational factors (e.g., aircraft performance, weight, speed), and human factors (e.g., pilot decision-making and situational awareness) [21, 14]. Despite extensive reporting through databases such as the **National Transportation Safety Board (NTSB) Aviation Accident database** and **Federal Aviation Administration (FAA) records**, there remain limited systematic analysis of which factors act as reliable precursors to runway excursions as well as how these factors interact.

The central problem addressed in this study is identifying candidate precursors to runway excursions using observational data and information-theoretic methods. Specifically, this study seeks to determine which operational, environmental, and human factors provide meaningful information about runway excursion risk as well as how these variables are statistically interdependent, without assuming true physical causation. By evaluating relationships among these variables, it becomes possible to prioritize safety interventions and inform predictive models for runway excursion prevention [24, 19, 6].

From this central problem, the following research questions are posed:

1. *Which measured operational, environmental, and human factors contain information about future runway-excursion risk beyond what is already explained by other variables?*

To address this question, this study evaluates the informational relationships between runway excursion events and a set of observed variables using mutual information and conditional mutual information. These measures quantify how much information provides about another variable while accounting for dependencies among multiple factors [1]. By estimating these quantities from empirical probability distributions derived from the available data, the analysis identifies variables that contribute meaningful predictive information regarding runway excursion risk. Variables that maintain non-zero conditional mutual information after accounting for additional observed factors are interpreted as candidate precursors to runway excursion events [22, 11, 25].

2. *What directed relationships between variables are supported by causation entropy analysis under reasonable modeling assumptions?*

This question is addressed by applying causation entropy analysis to infer directed informational relationships among the variables included in the dataset. Causation entropy extends conditional mutual information by evaluating whether a variable provides additional information about the target variable after conditioning on relevant factors [23]. In this framework, directional influence is inferred when a variable contributes statistically significant information beyond what is already explained by other variables within the system. Following the optimal causation entropy principle, parent sets are iteratively constructed for each target variable by selecting variables that maximize informational gain while accounting for previously selected predictors [23]. The resulting structure is interpreted as a directed dependency graph that highlights candidate precursor relationships among operational, environmental, and aircraft-related variables. Importantly, these relationships reflect statistical dependencies rather than confirmed physical causation, as the analysis is performed on observational data without controlled experimental design.

3. *How robust are inferred “precursors” across resampling, time windows, and estimator choices?*

This question evaluates the stability and reliability of the inferred informational relationships under different data deviations and modeling assumptions. Because mutual information and causation entropy estimates can be sensitive to sampling variability, discretization choices, and estimator selection, robustness is assessed through repeated sub-sampling of the dataset and evaluation across multiple temporal partitions. In addition, sensitivity analyses are conducted by varying estimation parameters and observing whether the same variables consistently emerge as significant contributors to conditional mutual information gain. Variables that remain stable across these deviations are interpreted as more reliable candidate precursors, while those that vary significantly across resamples are considered less robust or potentially context dependent. This step ensures that identified relationships are not artifacts of a specific sample configuration but reflect more persistent structures in the underlying data.

4. *Can weather volatility and runway operational conditions be used to predict high-risk scenarios for runway excursions, and which factors are most critical for intervention?*

This question focuses on identifying whether combinations of environmental and operational variables provide meaningful predictive structure for higher-risk runway excursion conditions. In particular, this question investigates whether variability in weather-related factors, such as visibility, wind speed, and precipitation, when combined with aircraft and operational characteristics, including aircraft make and model and Federal Aviation Regulations (FAR) operating requirements, is associated with increased severity or likelihood of runway excursion events. The analysis further seeks to determine which subset of variables contributes the most informational value toward distinguishing high-risk scenarios, with the goal of identifying practical indicators that could support early warning systems or operational decision-making. Rather than asserting deterministic causation, this question aims to isolate variables that consistently exhibit strong statistical dependence with adverse outcomes, making them potential candidates for further investigation in predictive safety modeling.

This study focuses on identifying statistical precursors and informational dependencies among observed variables associated with REs. The analysis is based on observational data and information-theoretic measures, which allow relationships between variables to be quantified without imposing strong parametric assumptions. While the results may highlight variables that provide meaningful predictive information about runway excursion risk, the findings of this study should be interpreted as identifying candidate causal structures rather than establishing definitive physical causation. These insights can help guide further investigation and support the development of data-driven safety strategies for mitigating runway excursion risk.

4 Background and Key Concepts

Information theory provides a mathematical framework for quantifying the relationships between random variables by measuring the amount of information they share. In the context of an aviation safety analysis, these information-theoretic tools enable researchers to measure how strongly operational, environmental, or

human factors variables are associated with runway excursion events. Rather than relying on parametric modeling assumptions, information-theoretic measures evaluate statistical dependencies directly from probability distributions. In this study, the primary quantities utilized are entropy, mutual information (MI), conditional mutual information (CMI), and causation entropy. When combined, these measures provide a systematic way in which to identify statistical relationships as well as potential precursors to aviation safety events.

One of the fundamental quantities in information theory is entropy, which measures the uncertainty or unpredictability of a random variable [1, 11]. For a discrete random variable X with probability distribution $p(x)$, Shannon entropy is defined as:

$$H(X) = - \sum_x p(x) \log p(x). \quad (1)$$

Entropy represents the expected amount of information contained in the outcome of a variable. Variables with many equally likely outcomes exhibit higher entropy, while variables with more predictable outcomes have lower entropy. This phenomenon can be illustrated by comparing a roulette wheel with a fair 6-sided dice [11]. A roulette wheel, represented by a random variable X , can take values 1 through 36, each with approximately equal probability, resulting in relatively high uncertainty. In contrast, although a fair six-sided die, represented by a random variable Y , also has equal probabilities for its six outcomes, the number of possible results is smaller, leading to lower overall uncertainty. In terms of entropy, the entropy of X is greater than the entropy of Y due to the level of uncertainty in the roulette wheel [11].

For aviation safety datasets, entropy can describe the variability of environmental conditions, operational parameters, or runway characteristics. High entropy indicates greater variability in observed conditions, while low entropy suggests more deterministic behavior in the variable being analyzed [1]. This study assumes that the variables analyzed are generated from a stationary process, which means that the statistical properties of the system do not change significantly over the period of time considered. Because the data set includes both continuous and categorical variables, continuous measurements including wind speed, precipitation, or visibility are discretized into categorical bins prior to estimating probability distributions. This approach enables entropy-based quantities to be estimated using empirical frequency distributions derived from the available data.

4.1 Mutual Information and Conditional Mutual Information

Mutual information (MI) is an information-theoretic measure that quantifies the amount of information shared between two random variables. It represents how much information is gained about one variable, X , after having observed a single value of another variable, Y [22, 11]. Formally, for discrete random variables X and Y , mutual information is defined as:

$$I(X; Y) = H(X) + H(Y) - H(X, Y), \quad (2)$$

where $H(X)$ and $H(Y)$ are the entropies of X and Y , and $H(X, Y)$ is their joint entropy [1]. MI is symmetric, meaning $I(X; Y) = I(Y; X)$ if the roles of X and Y are swapped; MI is also non-negative, with larger values indicating stronger statistical dependencies between variables [22]. In the context of runway excursions, mutual information can quantify how operational or environmental factors are associated with the occurrence of an event.

Conditional mutual information (CMI) extends the concept of mutual information by measuring the information shared between two variables X and Y after accounting for the effect of a third set of variables Z [25, 1]. Formally, CMI is defined as:

$$I(X; Y|Z) = H(X|Z) - H(X|Y, Z), \quad (3)$$

where $H(X|Z)$ is the entropy of X conditioned on Z , and $H(X|Y, Z)$ is the entropy of X conditioned on both Y and Z [1]. Intuitively, CMI quantifies the additional information that X provides about Y that is

not already explained by Z . This makes it particularly valuable in complex systems such as aviation safety, where multiple operational and environmental factors often interact.

Wyner (1978) extended the definition of CMI to arbitrary ensembles, providing a robust and general framework that applies beyond simple discrete variables [25]. This ensures that the measure remains valid in real-world datasets, which may include a mix of discrete and discretized continuous variables. In the context of this study, a variable X is considered a candidate precursor to runway excursions Y if it contributes non-zero CMI after conditioning on a set of other relevant factors Z . This approach enables a systematic assessment of the informational relationships among variables without relying on strict parametric assumptions [22, 11, 25].

4.2 Causation Entropy

Causation entropy is an information-theoretic measure used to identify directed relationships between variables by extending the concept of CMI. Whereas mutual information quantifies the amount of information shared between two variables and conditional mutual information measures that shared information after accounting for a conditioning set, causation entropy is specifically used to determine whether one variable provides a directed informational contribution to another beyond the influence of other relevant variables [23, 25, 1]. In complex systems such as aviation safety, where operational, environmental, and human factors may interact over time, a directional measure is useful for distinguishing variables that merely co-occur from those that may serve as candidate precursors to runway excursion events.

A useful precursor to causation entropy is *transfer entropy*, an information-theoretic measure used to quantify the information flow between two time-dependent processes [23]. In contrast to standard mutual information, which is symmetric and does not indicate direction, transfer entropy evaluates whether the past states of one process, Y , provides additional information about the future state of another process, X , beyond what is already contained in the history of process X [23]. In other words, transfer entropy measures whether knowledge of one variable's prior behavior reduces uncertainty about the future behavior of another variable in a directional sense. Formally, the transfer entropy from process $X^{(j)}$ to process $X^{(i)}$ is defined as:

$$T_{j \rightarrow i} \equiv h\left(X_{t+1}^{(i)} \mid X_t^{(i)}\right) - h\left(X_{t+1}^{(i)} \mid X_t^{(i)}, X_t^{(j)}\right) \quad (4)$$

This expression represents the reduction in uncertainty about the future value of $X^{(i)}$ when the past of $X^{(j)}$ is included in addition to the past of $X^{(i)}$. If the uncertainty is reduced, then the past of $X^{(j)}$ contributes information about the future of $X^{(i)}$, suggesting a directional informational relationship [23]. In the context of runway excursions, transfer entropy provides a conceptual way to examine whether earlier environmental or operational conditions contain information about later safety outcomes. However, because transfer entropy is typically evaluated pairwise, it may capture indirect influences rather than direct ones when multiple interacting variables are present [12, 23].

To address this limitation, causation entropy generalizes transfer entropy by allowing directional informational contributions to be evaluated conditional on a set of additional variables. Formally, causation entropy is defined as:

$$C_{J \rightarrow I \mid K} = h\left(X_{t+1}^{(I)} \mid X_t^{(K)}\right) - h\left(X_{t+1}^{(I)} \mid X_t^{(K)}, X_t^{(J)}\right) \quad (5)$$

This quantity can also be interpreted directly as conditional mutual information:

$$C_{X \rightarrow Y \mid Z} := I(X; Y \mid Z) \quad (6)$$

Under this interpretation, causation entropy measures whether a variable X provides directed informational contribution to a target variable Y after accounting for the explanatory effect of the conditioning set Z [25, 1, 23]. A non-zero value indicates that X contributes additional information about Y that is not already explained by the variables in Z . This makes causation entropy especially appropriate for aviation safety

analysis, where multiple conditions may jointly influence the occurrence of a runway excursion.

This study utilizes causation entropy to identify candidate precursor relationships among observed variables associated with runway excursions. A variable is considered a plausible precursor if it provides non-zero directed informational contribution to a target variable after conditioning on other relevant factors. Following the optimal causation entropy principle, the analysis proceeds by iteratively constructing a conditioning set for each target variable, selecting the variables that provide the most informative explanatory contribution, and then testing whether additional variables still provide unique information once that conditioning has been established [23]. This means building a set of parent variables that best explain a target variable and determining whether the inclusion of another candidate variable improves the explanation beyond what is already known.

The result of this analysis is a directed dependency structure, represented as either a parent set for each target variable or as a directed graph among variables. For this runway excursion analysis, this structure can help identify which environmental, operational, and human-factors variables appear to provide the strongest directed informational relationships with excursion-related outcomes. Importantly, these results should be interpreted as evidence of candidate causal structure rather than proof of true physical causation, since the analysis is based on observational data rather than controlled experimentation [12, 24].

Because this study includes both continuous and categorical variables, continuous measurements including wind speed, precipitation, visibility, and runway condition indicators will be discretized into categorical bins prior to estimating probability distributions. This enables causation entropy to be estimated consistently using empirical distributions derived from the available data, while maintaining interpretability across aviation safety variables [1, 22].

5 Data Description

Runway excursion (RE) events are analyzed using a merged observational dataset constructed from the National Transportation Safety Board (NTSB) Aviation Accident/Incident Database and historical METAR weather observations. At the industry level, runway excursions are defined as events in which an aircraft departs the runway surface during takeoff or landing, either by veer-off from the side of the runway or overrun beyond the end of the runway [5, 4]. To ensure consistency in event identification, this study classifies RE events using standardized aviation safety taxonomies. Event definitions align with the International Civil Aviation Organization (ICAO) and Commercial Aviation Safety Team (CAST) Common Taxonomy Team’s (CICCTT) occurrence category for RE, which distinguishes between landing (RE-L) and takeoff (RE-T) events [8]. This taxonomy is utilized as the conceptual basis for determining what qualifies as a runway excursion in the present analysis.

The dataset utilized within this study is event-based, meaning that the primary unit of analysis is an individual runway excursion event recorded in the NTSB database. Each event is associated with structured metadata describing the aircraft, location, operational context, and outcome severity [15]. To incorporate environmental context, each event is matched to METAR weather observations using airport proximity and timestamp alignment. Weather data is aggregated at the nearest hourly interval to each event time, forming a 24-hour environmental window centered around the event date, ensuring consistent temporal alignment between accident events and surrounding weather conditions.

A record is treated as a runway excursion if it satisfies at least one of the following criteria: (1) it is explicitly categorized as a runway excursion or equivalent occurrence type in the NTSB database, (2) it contains runway-excursion terminology in the event description or query filter, or (3) it can be mapped to the ICAO/CICCTT runway excursion taxonomy based on the event context [8, 4]. This definition is broad enough to capture both formally coded events as well as operationally relevant cases described in narrative form, while still remaining grounded in established industry terminology.

The final dataset contains 428 runway excursion events after preprocessing and merging. Each row within the dataset represents a single runway excursion event. Not every event contains complete contextual and operational data due to limitations associated with the NTSB’s role in foreign investigations. Under ICAO’s Annex 13, when an accident involving a US-registered civil aircraft occurs in a foreign state, the investigation is conducted by the state of occurrence, while the NTSB participates only as an accredited representative rather than the primary investigating authority [16]. As a result, the availability of detailed investigative information and final report materials is controlled by the foreign investigating authority, and not all cases provide complete or consistently structured data for inclusion in the public NTSB dataset [16, 7].

Event dates were converted to a standardized datetime format, and airport identifiers were cleaned and normalized prior to integration with METAR data. This standardization step ensured consistency across records and enabled reliable alignment between operational event data and environmental weather observations.

To supplement the event records with environmental context, hourly weather observations were collected from the Iowa State Mesonet ASOS/METAR archive, which provides archived weather observations reported at airport weather stations as well as historical METAR-format surface reports [9]. For each runway excursion event, the nearest weather station was identified using the airport code, and, when necessary, refined using geographic proximity based on the Haversine distance formula [20]. Using both methods, weather data was extrapolated for all 428 events in the NTSB dataset. METAR-based weather observations were retrieved for the corresponding event date.

These observations were then summarized at the hourly level across the 24-hour window centered on the event date in order to capture the environmental conditions surrounding each runway excursion. Once a 24-hour window was obtained for event dates, records were aligned using timestamp by matching closest weather observation to the event timestamp. Weather variables retained for analysis include temperature, dew point, relative humidity, wind speed, wind gust, visibility, altimeter setting, and one-hour precipitation, with selected present-weather indicators retained only if sufficiently complete.

5.1 Variables and Operational Definitions

Prior to data preprocessing, cleaning, and feature engineering, the original merged NTSB and METAR dataset contained 56 variables, providing operational and environmental context to runway excursion events. Variable columns were dropped from the merged dataset for several reasons. Duplicate columns, appearing in both the NTSB Accident/Incident data and the METAR weather data, were dropped to reduce redundancy within the data. Columns that were completely empty, or 95 percent of the column contained missing values were removed from the dataset, as these would not be beneficial to the analysis. Additionally, columns such as *report type* or *report status* were dropped from the dataset because they provided no meaningful information for runway excursion events, rather they indicated the type of NTSB report and the status of the report (all events in analysis were completed investigations) [15]. A further explanation of the data preprocessing is contained within the preprocessing of this analysis.

The table below summarizes all variables used in the cleaned NTSB and METAR dataset. Definitions follow the NTSB Aviation Accident/Incident Data Dictionary [15] as well as the Iowa State Mesonet METAR documentation [9]. Derived variables such as distance between event location and weather station and time difference between event occurrence and nearest METAR observation are computed using geographic and temporal alignment procedures described in the preprocessing pipeline. These derived features ensure consistent pairing between NTSB events and environmental conditions, especially when direct station alignment is not available.

Variable	Type	Unit	Description
ntsbno	Categorical	None	Unique NTSB event identifier for each accident/incident record.

Variable	Type	Unit	Description
eventtype	Categorical	None	Classification of event (Accident/Incident) per NTSB taxonomy.
eventdate	Datetime	UTC	Date and time of the aviation event.
city	Categorical	None	Geographic location of event occurrence.
state			
country			
n	Categorical	None	Internal identifier or auxiliary reporting field.
highestinjurylevel	Categorical	None	Most severe injury outcome (None, Minor, Serious, Fatal).
fatalinjurycount	Numerical	Count	Injury counts by severity level per event.
seriousinjurycount			
minorinjurycount			
probablecause	Categorical	None	Narrative cause description determined during investigation.
latitude	Numerical	Degrees	Geographic latitude of event location.
longitude	Numerical	Degrees	Geographic longitude of event location.
make	Categorical	None	Aircraft manufacturer involved in the event.
model	Categorical	None	Aircraft model involved in the event.
amateurbuilt	Categorical	Binary	Indicates whether aircraft was amateur-built.
numberofengines	Numerical	Count	Number of engines installed on aircraft.
purposeofflight	Categorical	None	Operational purpose (e.g., personal, training, commercial).
far	Categorical	None	FAA operational category (Part 91/121/135).
aircraftdamage	Categorical	None	Severity of aircraft damage (None, Minor, Substantial, Destroyed).
weathercondition	Categorical	None	Reported flight rules condition (IMC/VMC/Unknown).
operator	Categorical	None	Aircraft operator or organization.
docketurl	Categorical	None	Link to NTSB investigation report.
docketpublishdate	Datetime	UTC	Publication date of final report.
valid	Binary	None	Indicator of valid record after cleaning.
tmpf	Numerical	°F	Air temperature from METAR observation.
dwpf	Numerical	°F	Dew point temperature from METAR.
relh	Numerical	%	Relative humidity.
drct	Numerical	Degrees	Wind direction.
sknt	Numerical	knots	Wind speed.
gust	Numerical	knots	Wind gust speed.
vsby	Numerical	statute miles	Visibility at event time.
alti	Numerical	inHg	Altimeter pressure setting.
p01i	Numerical	inches	1-hour precipitation accumulation.
metar	Categorical	None	Raw METAR observation string.
weatherstation	Categorical	None	Closest ASOS/METAR station identifier.
weather_source	Categorical	None	Station assignment method (primary or fallback).
distance_nm	Numerical	nautical miles	Distance between event and station (Haversine-based).
time_diff	Numerical	hours	Time difference between event and matched METAR observation.
crosswind_component	Numerical	knots	Wind component perpendicular to runway axis (estimated using vector decomposition).
tailwind_component	Numerical	knots	Wind component aligned with runway direction; positive values indicate tailwind conditions.

Variable	Type	Unit	Description
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Table 1: Cleaned NTSB + METAR runway excursion dataset variables and definitions.

For analytical clarity, variables are grouped into three categories: METAR-based environmental variables, operational context variables, and aircraft characteristics. This categorization establishes a structured framework for feature selection and supports targeted investigation of potential precursor relationships, allowing environmental, operational, and aircraft-specific factors to be evaluated both independently and in combination.

METAR-based environmental variables capture the atmospheric conditions present at or near the time of each runway excursion event. These variables provide quantitative measures of weather-related factors that may influence aircraft performance, runway surface conditions, and pilot decision-making. By incorporating both directly observed METAR variables and derived wind components, this feature group enables a more precise representation of environmental context that may contribute to runway excursions.

Variable	Unit	Description
tmpf	°F	Air temperature at the time of the event.
dwpf	°F	Dew point temperature, indicating atmospheric moisture content.
relh	%	Relative humidity, representing the amount of moisture in the air.
drcr	Degrees	Wind direction relative to true north.
sknt	knots	Sustained wind speed.
gust	knots	Peak wind gust speed.
vsby	statute miles	Horizontal visibility at the time of the event.
alti	inHg	Altimeter setting, representing atmospheric pressure.
p01i	inches	One-hour precipitation accumulation.
crosswind_component	knots	Wind component perpendicular to runway heading (derived).
tailwind_component	knots	Wind component aligned with runway heading; positive values indicate tailwind (derived).

Table 2: METAR-based environmental variables used in runway excursion analysis.

Operational context variables describe the conditions under which each flight was conducted, including regulatory framework, flight purpose, and observed outcomes (highest level of injury obtained from runway excursion event). Operational variables provide insight into how different types of operations, such as commercial versus personal flights, may be associated with varying levels of risk. Additionally, categorical indicators such as flight rules and operator characteristics allow for the examination of how operational environments and organizational factors may influence the likelihood and severity of runway excursions.

Variable	Type	Description
weathercondition	Categorical	Flight rules condition at the time of the event (e.g., IMC, VMC, Unknown).
purposeofflight	Categorical	Operational purpose of the flight (e.g., personal, training, commercial).
highestinjurylevel	Categorical	Most severe injury outcome associated with the event.
far	Categorical	FAA operational category (e.g., Part 91, Part 121, Part 135).
operator	Categorical	Aircraft operator or organization responsible for the flight.

Table 3: Operational context variables describing flight conditions and usage.

Aircraft-specific variables capture characteristics related to the type and configuration of the aircraft involved in

each event. Aircraft-specific variables allow for differentiation between aircraft manufacturers and models, which may exhibit distinct performance capabilities, handling characteristics, and operational use cases. By including aircraft characteristics in the analysis, this allows for the assessment of whether certain types of aircraft are disproportionately represented in runway excursion events, and whether these differences may contribute to observed risk patterns.

Variable	Type	Description
make	Categorical	Aircraft manufacturer.
model	Categorical	Aircraft model designation.

Table 4: Aircraft-specific characteristics included in the analysis.

5.2 Missing Data and Data Quality

Missing data within the dataset arises primarily from incomplete NTSB investigative records, especially for foreign occurrence events under ICAO Annex 13 jurisdiction [16, 7]. Because the NTSB does not serve as the primary investigative authority in these cases, detailed operational and causal information is not always available in the public dataset.

To quantify missingness, the percentage of missing values was calculated for each variable following the initial merge of the NTSB and METAR datasets. Variables related to weather observations exhibited minimal overall missingness due to the use of station matching and fallback procedures based on geographic proximity. In contrast, several operational and investigative variables, including *probable cause*, *operator*, and *purpose of flight*, contained higher proportions of missing values. These gaps are not random, but instead reflect structural limitations in data reporting, particularly for international events.

A structured approach was implemented to address missing data depending on variable type and analytical relevance. For categorical variables, missing or blank entries within the dataset were standardized to explicit placeholders such as "UNKNOWN" or "NA". This approach preserves all 428 observations in the data while preventing the loss of potentially valuable events due to list-wise deletion. For example, *probable cause* and *weather condition* were assigned to "UNKNOWN" when missing, whereas *operator* and *FAR* were assigned "NA" where appropriate. By assigning placeholders, this ensures that missingness is explicitly represented as a meaningful category for categorical variables rather than implicitly ignored.

For numerical weather variables, missing values were handled using median imputation. As METAR observations were matched using nearest-station and nearest-time logic, missing values typically occur when a specific weather variable was not reported rather than due to complete data absence. Weather station-level median imputation was selected as a robust approach that minimizes the influence of outliers within the data while still preserving the overall distribution of the data. For the precipitation column, *p01i*, was assumed to be 0 if it was missing.

Variables with excessive missingness (greater than approximately 95 percent of entries in the dataset) were removed from the dataset during preprocessing, as they did not provide sufficient information for meaningful analysis. Additionally, duplicate or redundant variables arising from the dataset merge were eliminated to reduce dimensionality and avoid multicollinearity.

Outliers in numerical variables were assessed using distribution-based methods, including visual inspection of histograms and summary statistics. Rather than applying strict winsorization or clipping thresholds, outliers were retained in the dataset due to their potential relevance in aviation safety contexts. Extreme values in variables such as wind speed, visibility, or precipitation may correspond to hazardous environmental conditions that are directly related to runway excursion risk. Removing or artificially constraining these values could obscure meaningful relationships.

However, basic data cleaning steps were applied to ensure numerical stability. Infinite values and clearly invalid entries were removed or replaced with missing values prior to imputation. All numerical variables were also evaluated for reasonable physical bounds based on known aviation and meteorological standards. Overall, the missing data strategy prioritizes data retention while maintaining analytical integrity. By combining targeted imputation, explicit categorization of missing values, and removal of non-informative variables, the dataset remains both comprehensive and suitable for downstream statistical and information-theoretic analysis.

6 Data Exploration

To understand underlying relationships within the data, a descriptive overview of the dataset prior to any causal or information-theoretic analysis is given. This section summarizes the distributional properties, operational structure, and environmental conditions present in the final cleaned dataset of 428 runway excursion events. The goal is to characterize how the data behaves prior to introducing any causal or predictive modeling assumptions.

Before interpreting the dataset, it is important to define key aviation safety terminology used in this analysis. Formal definitions of *aircraft accident*, *aircraft incident*, *fatal injury*, *serious injury*, and *substantial damage* are provided under 49 CFR Part 830 (NTSB reporting regulations) [3]. An aircraft accident is defined as an occurrence associated with the operation of an aircraft what results in fatal or serious injury, or substantial damage to the aircraft. An aircraft incident refers to an occurrence in which the threshold for an accident is not met but still affects or could affect safety of operations. These definitions are critical for interpreting outcome variables contained in this dataset, as they define how events are classified within the NTSB reporting framework.

6.1 Event Type and Regulatory Classification

The distribution of event types is shown in Table 5. The dataset is dominated by accident events (ACC), with relatively few incidents. This imbalance is expected given that the dataset was constructed around runway excursion events that meet NTSB reporting thresholds and contain sufficient documentation for analysis [3]. Table 5 establishes that most observations are classified as accidents rather than incidents, which is most important when interpreting downstream results, as the dataset is not representative of all aviation safety occurrences but rather those with higher reporting severity.

Event Type	Count
ACC	399
INC	29

Table 5: Distribution of event types in the dataset.

Regulatory context for runway excursions is summarized in Table 6, which presents FAA FAR classifications. Pat 91 operations (091) heavily dominate the dataset, indicating that most runway excursion events within the data occur in general aviation rather than commercial airline operations. Smaller proportions of Part 135 and Part 137 operations suggest limited representation of charter and agricultural operations. This distribution is important because operational environment strongly influences both risk exposure and reporting structure, and therefore acts as a key contextual variable in later analysis.

FAR Category	Count
091	358
135	16
NUSN	12
NUSC	9
137	8

Table 6: Distribution of FAA regulatory categories (FAR codes).

6.2 Purpose of Flight (Operational Context)

The operational context of each flight is further described by purpose of flight, as shown in Table 7. Personal (PERS) flights represent the majority of events, followed by (INST) flights. A notable proportion of entries within the dataset are categorized as UNKNOWN, reflecting missing or incomplete reporting within the NTSB dataset. This distribution reinforces the strong representation of general aviation activity in runway excursion events. The presence of instructional flights is also operationally relevant, as training environments often involve lower experience pilots and higher variability in decision-making conditions.

Purpose	Count
PERS	259
INST	64
UNKNOWN	46
BUS	18
POSI	10

Table 7: Distribution of purpose of flight categories.

6.3 Outcome Severity Distribution

Outcome severity of runway excursions is summarized in Tables 8 and 9, where Table 8 shows the distribution of injury severity levels across all events, and Table 9 presents the distribution of aircraft damage severity. The majority of events result in no injuries (81.3 percent), with minor injuries accounting for 11.7 percent, and serious and fatal outcomes representing a relatively small fraction of the dataset. This strong imbalance in injury distribution indicates that most general aviation runway excursions do not result in severe human injury, which is important when interpreting later modeling results, particularly in terms of class imbalance and predictive bias.

Injury Level	Proportion
None	0.813
Minor	0.117
Fatal	0.040
Serious	0.030

Table 8: Distribution of highest injury level across all events.

Similar to injury outcome, the dataset is heavily skewed toward less extreme aircraft damage outcomes, with 88.8 percent of events resulting in substantial damage and only a small proportion resulting in minor or destroyed classifications. This suggests that while injuries are relatively uncommon, aircraft damage is frequently significant, highlighting the operational importance of runway excursion events even when they are not fatal.

Damage Type	Proportion
Substantial	0.888
Minor	0.058
Destroyed	0.037
NA	0.016

Table 9: Distribution of aircraft damage severity.

To further understand how human injury outcomes relate to physical aircraft damage, Figure 1 visualizes injury severity conditioned on aircraft damage category. The chart illustrates how injury severity varies across different levels of structural damage and shows that the relationship is not strictly linear. In events classified as having minor damage or where damage is recorded as NA, all observed outcomes correspond to no injuries, which is consistent with low-energy or low-impact occurrences. For substantial damage events, no-injury outcomes dominate, though small proportions of minor, serious, and fatal injuries are present, indicating that significant structural damage can still occur without severe occupant harm.

The most notable variation appears in the destroyed category, where injury outcomes span the full range from no injury to fatal injury. This reflects the variability in aircraft crash dynamics, where survivability depends on factors such as impact forces, crash geometry, fire involvement, and restraint effectiveness. While injury severity increases with aircraft damage, damage level alone is not a deterministic predictor of human outcome.

6.4 Aircraft Characteristics

Aircraft manufacturer distribution is summarized in Table 10. The dataset is dominated by a small number of general aviation manufacturers, with Cessna and Piper accounting for the majority of recorded runway excursion

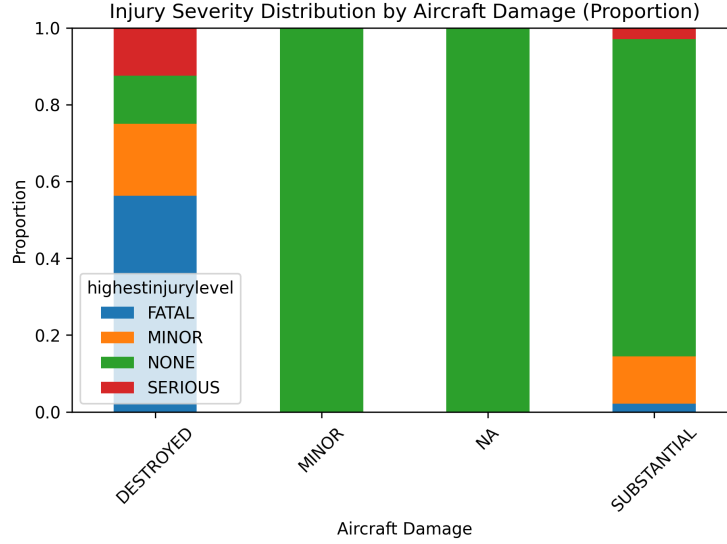


Figure 1: Injury severity proportioned to aircraft damage.

events. This concentration is consistent with the overall structure of the dataset, which primarily reflects general aviation operations rather than commercial airline activity. Smaller contributions from manufacturers such as Beech, Boeing, and Mooney indicate limited representation of larger transport-category and specialized aircraft within the dataset.

Aircraft Make	Count
CESSNA	120
PIPER	79
BEECH	21
BOEING	11
MOONEY	10

Table 10: Most frequent aircraft manufacturers in the dataset.

6.5 Weather Conditions

Weather conditions at the time of runway excursion events are summarized in Table 11. The dataset is overwhelmingly dominated by Visual Meteorological Conditions (VMC), with a small number of Instrument Meteorological Conditions (IMC) cases and a modest proportion of UNKNOWN entries. This distribution indicates that runway excursion events are not primarily driven by low-visibility or IMC, but rather frequently occur under visually favorable conditions as well. This suggests that environmental visibility alone is not sufficient to explain runway excursion occurrence. Runway excursions appear to arise under a wide range of meteorological conditions, reinforcing the need to consider additional environmental and operational factors.

Weather Condition	Count
VMC	379
UNKNOWN	28
IMC	21

Table 11: Distribution of flight weather conditions.

To further examine environmental conditions, wind speed distributions were analyzed. Figure 2 illustrates the distribution of wind speeds (in knots) at the time of runway excursion events. The histogram reveals a strong

concentration of observations in the 0-5 knot range, with frequencies decreasing steadily as wind speed increases. This produces a right-skewed distribution with a long tail extending towards higher wind speeds.

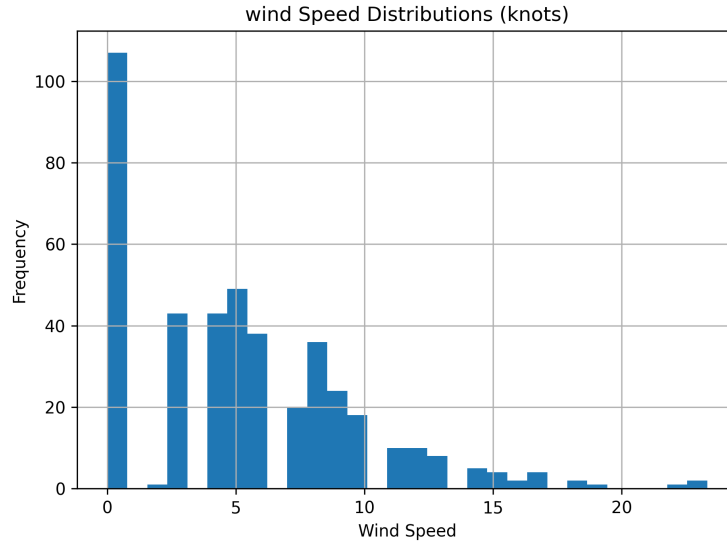


Figure 2: Distribution of wind speed (knots) for runway excursion events.

This pattern suggests that most runway excursion events occur in relatively calm or light wind conditions rather than during high-wind situations. While strong winds are known to contribute directional control difficulties and unstable approach or landing conditions, the observed distributed indicates that wind speed alone is not a dominant factor in the occurrence of runway excursion events within this dataset. Instead, the concentration of events under low wind conditions implies that other variables, such as crosswind components, aircraft configuration, or operational decision-making may play a more significant role in contributing to excursion risk.

To further characterize wind behavior during runway excursion events, Figure 3 illustrates the relationship between crosswind and tailwind components derived from wind speed and direction. This decomposition represents how the wind vector projects onto axes aligned with the runway, separating lateral (crosswind) and longitudinal (tailwind/headwind) effects.

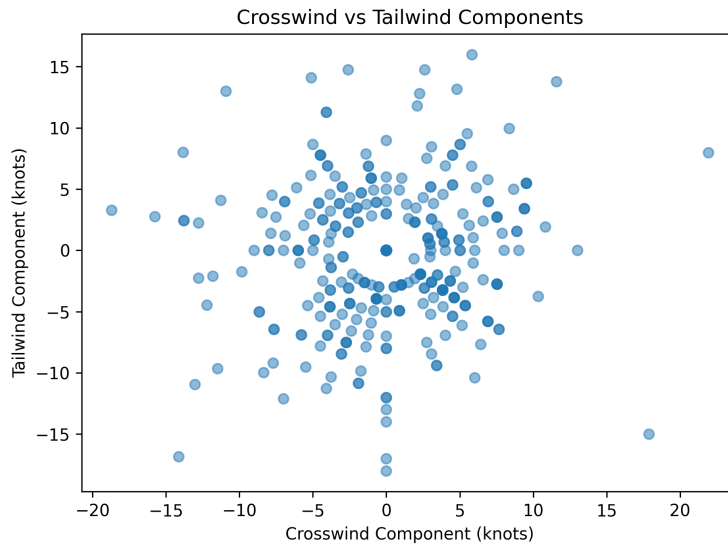


Figure 3: Distribution of crosswind and tailwind components.

The plot shows a dense concentration of observations near the origin, indicating that most runway excursion events occur under relatively light wind conditions, where both crosswind and tailwind magnitudes are small. The overall distribution is approximately symmetric across quadrants, which indicates there is no dominant directional bias in wind orientation relative to runway alignment. As a result, headwind, tailwind, left crosswind, and right crosswind conditions all appear with comparable frequency across the dataset.

Each point in the visualization represents the relative contribution of wind acting along and across the runway axis. While extreme values are present, they are sparse, reinforcing the notion that most operational conditions fall within moderate and manageable wind envelopes. However, the spread across all quadrants highlights that aircraft are exposed to a wide range of directional wind components, which may contribute differently to runway excursion risk depending on runway orientation, aircraft type, and operational context.

6.6 METAR Time Alignment Quality

The quality of temporal alignment between runway excursion events and METAR observations is summarized in Table 12. The median time difference between an event and its matched weather observation is only 10 minutes, while nearly 90 percent of events are aligned within one hour. Although the mean time difference is larger due to a small number of outliers, the median and percentile-based measures indicate strong overall alignment. This level of temporal consistency supports the validity of linking environmental conditions to operational events, ensuring that the extracted weather variables reasonably represent conditions at or near the time of each runway excursion.

Metric	Value
Mean Time Difference	2 days 01:48:13
Median Time Difference	10 minutes
Within 1 hour	89.95%
Within 30 minutes	88.79%

Table 12: Quality of METAR-event time alignment.

6.7 Distributional Structure

To better understand the statistical structure of environmental variables used in this analysis, distributional behavior was analyzed using histograms and log-transformed representations. The four histograms in Figure 4 illustrate the effect of logarithmic transformations on key weather variables associated with runway excursion events. These transformations reduce skewness and improve interpretability by compressing extreme values while preserving overall structure.

Wind speed ($\log(1 + \text{sknt})$) is heavily concentrated near zero, reflecting that most events occur under light wind conditions, with a long right tail representing infrequent high-wind cases. Gust ($\log(1 + \text{gust})$) shows a similar right-skewed structure, but with a sharper peak at moderate transformed values, indicating that gusty conditions are present but unevenly distributed in raw scale. Visibility ($\log(1 + \text{vsby})$) is tightly clustered, suggesting that most events occur under moderate visibility conditions rather than extreme low-visibility scenarios. Precipitation ($\log(1 + \text{p01i})$) is dominated by a spike near zero, confirming that measurable precipitation is relatively rare at event times, even though runway contamination may still occur without active rainfall.

6.8 Correlation/Association Overview

The correlation heatmap in Figure 5 summarizes pairwise linear relationships among numeric and engineered features. While it provides slight insight into variable structure, it is important to emphasize that correlation among variables does not imply causation, particularly in observational aviation safety data where many confounding operational factors are present.

Injury-related variables (fatal, serious, minor) exhibit moderate positive correlations, reflecting that higher severity events often involve multiple injury categories. Meteorological variables show physically consistent relationships: *temperature* (*tmpf*), *dew point* (*dupf*), and *relative humidity* (*relh*) are interrelated, while wind variables (*drct*, *sknt*, *gust*, *crosswind*, and *tailwind*) form a coherent cluster driven by shared atmospheric structure.

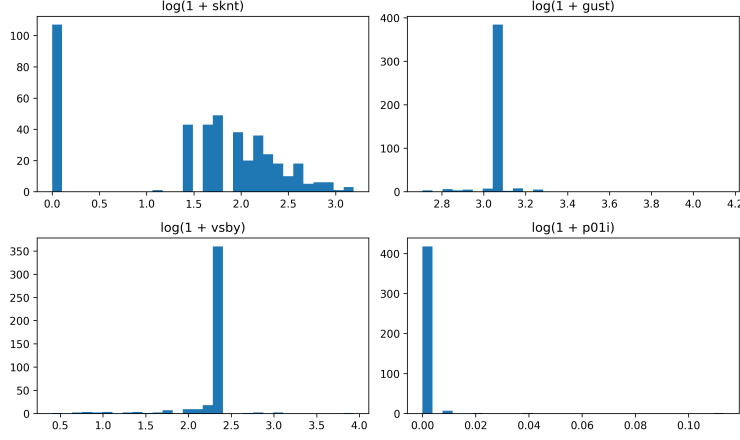


Figure 4: Log-transformed distributions of key meteorological variables.

Engineered wind components show expected dependence on wind direction and wind speed but remain weakly correlated with injury outcomes. Similarly, geographic variables (latitude and longitude) as well as operational descriptors (e.g., number of engines, distance to weather station) show limited linear association with most other variables.

The binary damage outcome exhibits only weak correlations with both injury severity and environmental conditions, reinforcing that runway excursions are driven by a combination of interacting factors rather than single-variable effects. The heatmap is used as a structural diagnostic tool rather than a causal indicator, helping identify redundancy and potential feature groupings for later information-theoretic analysis.

6.9 Feature Selection Summary

Feature selection was performed using mutual information (MI) and conditional mutual information (CMI) to identify variables most strongly associated with runway excursion outcomes. MI captures nonlinear marginal dependencies, while CMI accounts for redundancy between variables and isolates incremental information.

The mutual information approaches (custom and sklearn-based) produced largely consistent results, identifying both operational and environmental variables as primary contributors. Features such as *weather condition* (VMC and UNKNOWN), *purpose of flight* (particularly UNKNOWN), and *aircraft make* (e.g., GULFSTREAM) appeared consistently across MI rankings, suggesting strong marginal associations with the outcome variable.

Conditional mutual information refined these results by selecting variables that provide unique explanatory value beyond already selected features. The greedy CMI selection process identified *purpose of flight* (UNKNOWN), *wind direction* (*drct*), *altimeter setting* (*alti*), *temperature* (*ttmpf*), and *dew point* (*dwpf*) as the most informative subset. These variables capture a combination of operational uncertainty and environmental conditions, indicating that both domains contribute independently to runway excursion outcomes.

Comparison between MI and CMI results reveals a small set of overlapping features, including *purpose of flight* (UNKNOWN) and *weather condition*, which appear robust across both methods. These overlapping variables are likely strong predictors due to their consistent presence regardless of whether redundancy is considered. The feature selection results suggest that runway excursion risk is influenced by a combination of environmental conditions as well as operational context rather than any single dominant factor.

7 Methodology

An information-theoretic workflow is used in this study to identify precursor relationships among runway excursion event characteristics and matched historical weather conditions. The overall methodology is designed to move from raw observational data to a structured set of variables suitable for mutual information, conditional mutual information, and causation entropy analysis. Because the source data are heterogeneous as well as partially incomplete, the methodology places substantial emphasis on preprocessing, feature consistency, and estimator stability.

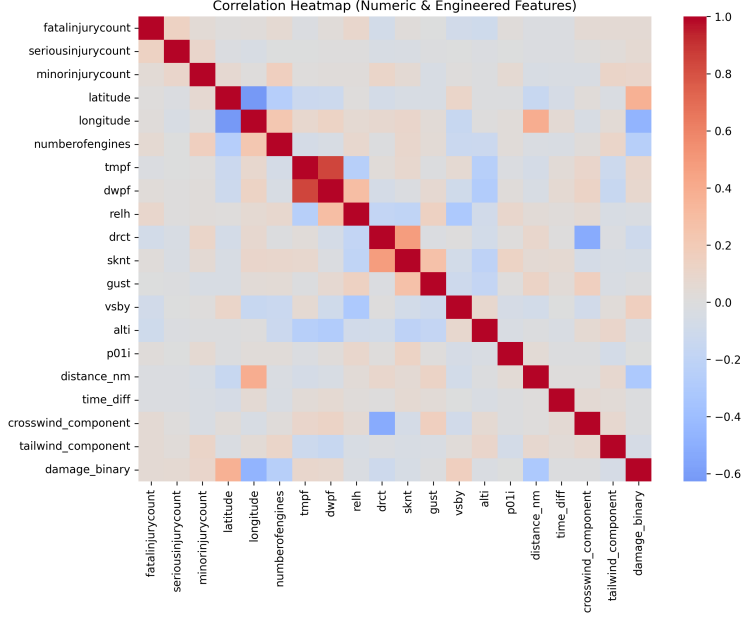


Figure 5: Correlation heatmap of numeric and engineered variables.

The analysis is implemented in Python using a reproducible, modular workflow. The project repository is organized into separate components for raw data ingestion, cleaning, weather-event matching, feature construction, exploratory analysis, and information-theoretic modeling, as this structure helps ensure that each stage of the project workflow can be independently verified and reproduced. Key libraries used in this analysis include pandas, NumPy, scipy, matplotlib, scikit-learn, and NetworkX, along with custom implementations of mutual information (MI) and conditional mutual information (CMI) estimators developed for this project.

The methodology consists of five stages:

1. preprocessing and data cleaning,
2. feature construction and encoding,
3. estimation of entropy-based quantities,
4. causal discovery using causation entropy, and
5. robustness and significance assessment.

7.1 Preprocessing Pipeline

In order to begin the preprocessing pipeline, METAR data had to be obtained from the Iowa State Mesonet ASOS/METAR archive [9]. These data provide hourly weather observations recorded at airport weather stations and serve as the environmental component of the merged dataset. Airport identifiers in the NTSB dataset were first standardized from FAA/IATA format to ICAO format to ensure compatibility with METAR reporting. Hourly weather data corresponding to each event date were then retrieved for each airport identifier. Initial retrieval successfully matched weather observations for 285 events, while 143 events did not have a direct station match.

To address missing station coverage, a nearest-station approach was implemented using geographic proximity. Airport locations were matched to the closest available ASOS weather station using the Haversine distance formula, which calculates the distance between two points on the Earth’s surface by first computing Euclidean distance between these points and then converting this distance to one measured along the Earth’s surface using spherical coordinates [20]. Station metadata were obtained from the National Weather Service ASOS station listing [17]. This approach allowed for full weather coverage across all 428 runway excursion events.

Once weather observations were obtained, a 24-hour window of METAR data was constructed for each event, with

over 14,000 total weather observations. These observations were aligned temporally by selecting the METAR report closest to the event timestamp. All other observations within the window were removed, resulting in a single matched weather observation per runway excursion event. The METAR dataset was then merged with the NTSB dataset using the unique event identifier.

Following data integration, cleaning procedures were applied. Columns were removed from the dataset if they were redundant, irrelevant to the analysis, or contained excessive missing data, greater than 95 percent. This included columns such as report number, report status, weather codes, station identifiers, and docket publication date. Duplicate columns present in both datasets, such as event identifiers and airport identifiers, were also removed to reduce redundancy. Data types were then standardized to ensure consistency across variables. Numerical variables were converted to numeric format, categorical variables were retained as strings, and datetime variables were converted to UTC-aware datetime objects utilizing pandas. This ensured consistency across the dataset and enabled reliable temporal comparisons.

Missing data were addressed based on variable type and context in relation to runway excursions. For categorical variables describing accident or operational characteristics, such as probable cause, weather condition, and purpose of flight, missing values were replaced with "UNKNOWN" to preserve observations while indicating incomplete information. For administrative or non-operational categorical fields, including operator, state, docket URL, or FAR operational category, missing values were standardized as "NA". Weather variables with partial missingness were imputed using station-level median values to maintain local environmental consistency. Precipitation (p01i) was assumed to be zero when missing, reflecting the absence of recorded precipitation in METAR reporting [9].

Following data cleaning, two additional features were engineered to provide more meaningful environmental context. Crosswind and tailwind components were calculated using wind speed and direction variables for runway excursion events, representing wind components perpendicular and parallel to the runway axis, respectively. These derived variables provide more operationally relevant measures of wind impact compared to raw wind observations. The final cleaned dataset, consisting of 428 runway excursion events with complete weather coverage, was saved for downstream analysis. Exploratory data analysis (EDA), visualization, and feature selection (custom MI, sklearn MI, custom CMI) were performed to identify candidate precursor variables.

7.2 Estimating CMI

To quantify relationships between variables beyond simple pairwise dependence, conditional mutual information (CMI) is used to measure the information shared between a candidate predictor and the target variable while conditioning on an existing set of selected variables, which enables the analysis to distinguish between variables that provide unique explanatory power and those that are redundant given other features [2]. In this analysis, the target variable was constructed as a binary indicator of runway excursion severity. The original *aircraft damage* variable was transformed into *damage.binary*, where events classified as "Substantial" or "Destroyed" were assigned a value of 1, and all other outcomes, including "Minor" and "NA", were assigned a value of 0. This formulation captures operationally significant outcome while reducing sparsity in rare categories and supporting more stable estimation.

Because the dataset contains a mixture of categorical and continuous variables, a discretization-based approach was used for entropy and mutual information estimation. Continuous variables, including METAR-derived weather features were discretized using equal-frequency (quantile) binning. This approach ensures that each bin contains approximately the same number of observations, which is particularly important given the relatively small sample size of 428 events. In practice, variables were discretized into between 5 and 10 bins, and sensitivity analysis was performed to confirm that feature rankings remained stable across different bin counts.

Categorical variables were retained in their original form, including explicit placeholders such as "UNKNOWN" and "NA", which were preserved to represent missing or incomplete information. This allows the estimator to capture potential signal associated with reporting gaps rather than discarding those observations.

Entropy, mutual information (MI), and conditional mutual information (CMI) were estimated using empirical probability distributions derived from discretized data. Recall that for a discrete variable X , entropy is computed as:

$$H(X) = - \sum_x p(x) \log p(x) \quad (7)$$

and conditional mutual information is defined as:

$$I(X; Y|Z) = H(X|Z) - H(X|Y, Z). \quad (8)$$

These quantities were computed using frequency-based estimators over joint and conditional distributions [2, 22].

A greedy forward-selection procedure was then applied for feature selection. At each iteration, the variable that maximized $I(X; Y|S)$, where S represents the set of already-selected variables, was added to the model. This process continues until additional variables provide negligible incremental information, resulting in a compact set of non-redundant predictors.

Estimator stability was evaluated through several robustness checks. First, sensitivity to discretization was assessed by varying the number of bins and confirming that the ranking of top features remained consistent. Second, bootstrap resampling was utilized to repeatedly estimate MI and CMI across multiple subsamples of the dataset, allowing variability in feature importance to be assessed. Third, results from the custom MI implementation were compared against those obtained using scikit-learn’s mutual information estimator to ensure consistency across methods.

7.3 Causal Discovery via Causation Entropy

Causal discovery is performed using a greedy forward selection procedure based on causation entropy (CSE), which is closely related to conditional mutual information and reduces to it under specific conditions, and captures the incremental dependence between a candidate variable and the target after conditioning on previously selected variables [2, 23]. The objective of causal discovery is to identify a compact subset of variables that maximizes information about the runway excursion outcome while removing redundancy through conditioning.

The procedure begins by choosing a target variable Y , defined as the binary runway excursion severity indicator (`damage_binary`), where a value of 1 corresponds to substantial or destroyed aircraft damage and 0 corresponds to all other outcomes (minor, none, or NA). The conditioning set is initialized as empty, $Z = \emptyset$, and the candidate variables are evaluated individually based on their mutual information shared with the target variable.

At each iteration, the algorithm computes the conditional mutual information $I(X; Y|Z)$ for all remaining candidate variables X . The variable that maximizes this quantity is selected and added to the conditioning set, updating it as $Z \leftarrow Z \cup \{X^*\}$. This process is repeated sequentially, allowing each newly added variable to be evaluated in the presence of those already selecting, which ensures that only incremental information is retained.

The selection process follows a greedy forward-selection procedure constrained primarily by a predefined maximum number of features (`max_features` = 6). At each iteration, the feature that maximizes CSE is selected and removed from the candidate pool. The process terminates once the maximum feature limit is reached or if no candidate feature produces a positive improvement in the objective function. Statistical significances is not used as a stopping condition during selection, but instead is evaluated using permutation test, where the target variable is randomly permuted to generate a null distribution of causation entropy values. A feature is retained only if its observed value exceeds what would be expected under this null distribution, ensuring that selected variables reflect non-random structure within the data.

No explicit backward pruning step is applied. Instead, redundancy is controlled through the conditioning structure itself, as variables that do not contribute additional information beyond those already selected will naturally yield low incremental CSE and will not be chosen in subsequent iterations.

7.4 Significance Testing and Robustness

To reduce the risk of over-interpreting noise-driven dependencies, multiple complementary robustness procedures are implemented in the analysis.

Permutation testing is used in this study to evaluate whether observed dependencies between target variables and the target outcome are statistically meaningful or could plausibly arise under a null hypothesis of no association. The method works by randomly permuting the target variable, $Y = \text{damage_binary}$, while keeping the predictor variables fixed, breaking any true dependency structure while preserving the marginal distributions of the data [18]. The information-theoretic statistic of interest is then recomputed under repeated permutations to generate an empirical null distribution. The observed value is compared against this distribution, and statistical significance is assessed based on how extreme the observed statistic is relative to the permuted outcomes. This approach does not rely on parametric assumptions and is particularly well-suited for nonlinear dependence measures such as mutual information

and causation entropy [18].

Within this analysis, permutation testing is applied after feature selection to evaluate whether the observed mutual information, conditional mutual information, and causation entropy values are statistically significant relative to a null distribution. For each feature-target pair, the target variable is randomly permuted multiple times, and the corresponding dependency measure is recomputed for each permutation. The resulting distribution represents the expected behavior of the statistic under independence. A p-value is computed as the proportion of permuted statistics that are greater or equal to the observed value. Features that consistently exceed the null distribution are treated as statistically significant and are retained during the forward selection process.

In addition to permutation testing, bootstrap stability is used as a complementary robustness assessment to evaluate how sensitive the feature selection process is to sampling variability. In this approach, the dataset is repeatedly resampled with replacement, and the full causation entropy-based forward selection procedure is rerun on each bootstrap sample. This produces a distribution over selected features across many resampled datasets, allowing for an empirical measure of how consistently each feature is selected.

The frequency with which a feature appears in the selected set across bootstrap iterations is used as a measure of stability. Features that are consistently selected are interpreted as robust precursors, while those that appear inconsistently are more sensitive to noise or disturbances in the data. This procedure follows standard bootstrap-based inference principles for evaluating estimator stability in complex statistical models [26]. A variable is considered a robust precursor if it demonstrates both statistical significance under permutation testing and high selection stability under bootstrap resampling, ensuring that identified relationships are not artifacts of noise or finite sample variation.

8 Results

Feature selection was performed using mutual information (MI), conditional mutual information (CMI), and causation entropy (CSE). These measures capture non linear dependence structures between environmental, operational, and aircraft-level variables and the runway excursion outcome. The results of this analysis are presented in terms of both ranked feature importance and the inferred dependency graph structure derived from the forward selection procedure.

The results are interpreted as associative structure learned under conditional independence assumptions rather than strict causality. All dependencies are therefor reported as conditional relationships supported by information-theoretic evidence under the modeling assumptions of the framework.

Additionally, the results of this analysis are organized to directly address the research questions defined in Section 3. First, variables that contain unique informational value about runway excursion outcomes are identified (Research Question 1). Next, the directed dependency structure inferred through causation entropy is presented (Research Question 2). Finally, the stability of these relationships is evaluated under resampling and perturbation (Research Question 3), along with a discussion of predictive structure in relation to environmental and operational conditions (Research Question 4).

8.1 Informational Precursors to Runway Excursions

This section addresses Research Question 1 by identifying which variables contain unique informational value regarding runway excursion outcomes beyond what is explained by other observed variables. Feature selection was performed using mutual information and conditional mutual information, where variables that maintain non-zero conditional dependence with the target after accounting for previously selected variables are interpreted as candidate precursors.

The results indicate that several operational and environmental variables retain measurable informational contribution under conditioning. In particular, *wind direction (drct)*, *altimeter setting (alti)*, *FAR classification (far)*, and *purpose of flight (purposeofflight)* consistently appear as high-information contributors in the selection process. These variables remain informative even after accounting for redundancy among other predictors, suggesting that they provide distinct signals related to runway excursion outcomes. The variable *wind speed (sknt)* shows weaker but still non-negligible contribution, while *dew point (dwpf)* does not maintain strong conditional information once other variables are included.

Feature	CSE Score	p_value	Stability
drct	0.2681	0.0000	1.0000
alti	0.2665	0.8000	1.0000
far	0.2105	0.0000	0.9200
purposeofflight	0.1797	0.0000	0.8000
sknt	0.1524	0.3400	0.3800
dwpf	0.0000	0.4900	0.2800

Table 13: Causation Entropy Feature Selection Results

Table 13 summarizes the causation feature selection results, where higher values indicate stronger conditional information contribution to the target variable. The ranking aligns with the qualitative interpretation above. The variables *drct* and *alti* emerge as the strongest contributors, followed by *far* and *purposeofflight*, while *sknt* and *dwpf* exhibit comparatively weaker conditional dependence. Together, these results highlight a small set of variables that retain unique informational value regarding runway excursion severity after accounting for dependencies among the broader predictor set.

8.2 Causal Graph / Network View

To address Research Question 2, this section presents the directed dependency structure inferred through causation entropy analysis. Each directed edge represents a variable that provides statistically significant additional information about the runway excursion outcome after conditioning on previously selected variables. Edge weights correspond to estimated causation entropy values, and should be interpreted as conditional informational strength rather than physical causal effect sizes.

The resulting structure can be represented as a directed dependency graph (digraph), where nodes correspond to variables and directed edges indicate conditional information flow from a source variable to the target [10]. In this framework, a directed edge from variable *X* to the target variable *Y* indicates that *X* contributes additional information about the outcome beyond what is explained by other variables in the conditioning set. Directed graphs are commonly used to represent structured relationships in complex systems, where edges encode directional dependencies between variables [10].

The inferred graph exhibits an interconnected network among both environmental and operational variables. Multiple variables sever as both sources and receivers of information, indicating that dependencies are not limited to direct relationships with the runway excursion outcome, but also exist among predictors themselves.

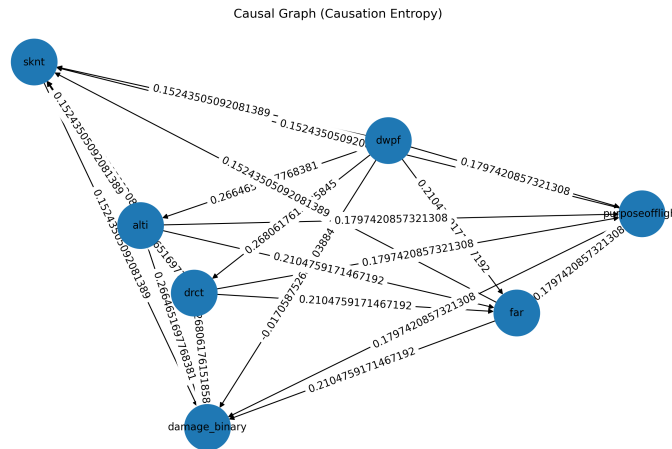


Figure 6: Causal graph derived from causation entropy analysis. Directed edges represent conditional dependencies between variables, with edge weights corresponding to estimated causation entropy values.

Several structural patterns are apparent in the graph. Environmental variables, including *wind speed (sknt)*, *wind direction (drct)*, *dew point (dwpf)*, and *altimeter setting (alti)*, form a densely connected sub-network. These variables exchange conditional information with one another, reflecting underlying physical relationships among atmospheric conditions. For example, wind-related variables and pressure-based measurements are inherently linked through shared meteorological processes, which is reflected in the multiple directed edges among these nodes.

Operational variables, including *purposeofflight* and *far*, appear downstream of several environmental and intermediate variables. These nodes receive directed edges from multiple sources, suggesting that their informational relationship with runway excursion outcomes may be influenced by both environmental context and operational categorization. In particular, *far* appears to aggregate information from multiple upstream variables before contributing to the target variable, indicating that regulatory classification may reflect a combination of underlying operational and environmental conditions.

The target variable, **damage.binary**, is positioned as a terminal node that receives directed edges from several predictors, including *drct*, *alti*, *far*, and *sknt*. These edges represent variables that provide incremental information regarding runway excursion severity after accounting for dependencies among other predictors. Among these, *drct* and *alti* exhibit the strongest edge weights, indicating that wind direction and atmospheric pressure provide the most substantial conditional informational contribution to the outcome.

The presence of intermediate dependencies among predictors suggests that some variables may influence the target indirectly through other variables. As an example, environmental variables such as *dwpf* and *sknt* contribute information to other predictors, which in turn contribute to the target variable. This layered structure highlights the possibility of indirect pathways of influence, rather than purely independent direct effects.

source	target	weight	p_value	type
dwpf	damage.binary	0.0000	0.4900	conditional_dependency
drct	damage.binary	0.2681	0.0000	conditional_dependency
alti	damage.binary	0.2665	0.8000	conditional_dependency
far	damage.binary	0.2105	0.0000	conditional_dependency
purposeofflight	damage.binary	0.1797	0.0000	conditional_dependency
sknt	damage.binary	0.1524	0.3400	conditional_dependency

Table 14: Causal Graph Edge List Derived From Causation Entropy

Table 14 summarizes the list of edge weights derived from causation entropy procedure and provides a numerical view of the conditional dependencies identified in the graph. The table reinforces several patterns visible in the network representation. Variables such as *drct*, *alti*, and *far* exhibit comparatively higher causation entropy values when directed toward the target, indicating stronger conditional informational contributions, whereas others such as *dwpf* and *sknt* show weaker or statistically insignificant effects once conditioning is applied. Together, the graph and the accompanying edge list provide a consistent depiction of a system in which both environment and operational variables contribute layered, interdependent information about runway excursion severity, with no single predictor dominating the dependency structure.

The inferred graph supports a structured interpretation of runway excursion risk in which both operational and environmental variables contribute independent informational signals. The presence of dependencies among predictors suggests that runway excursion outcomes arise from interacting factors rather than isolated variables. However, because the analysis is based on observational data and information-theoretic measures, these directed edges should be interpreted as conditional dependencies supported by the data under the modeling assumptions, rather than definitive causal mechanisms.

8.3 Sensitivity Analyses

To address Research Question 3, this section evaluates the robustness of the inferred precursor relationships under different estimator choices, discretization schemes, as well as resampling procedures. Research Question 4 is also addressed in this section by examining whether stable combinations of environmental and operational variables provide meaningful structure for identifying higher-risk runway excursion conditions.

To assess sensitivity to estimator parameters, the number of bins used for discretizing continuous variables was varied within a range of 5 to 10 bins. Across these configurations, the relative ranking of the top features remained largely consistent. In particular, *drct*, *alti*, and *far* consistently appeared among the highest-ranking variables regardless of discretization choice. This suggests that the informational contribution of these variables is not strongly dependent on binning strategy and reflects stable structure in the data rather than artifacts of discretization.

Additional sensitivity analysis was performed by removing rare or extreme observations, particularly events associated with unusually high wind speeds or large time gaps between METAR observations and NTSB event timestamps. The removal of these cases resulted in minor changes to lower-ranked features but did not substantially alter the ordering of the primary variables. The persistence of *drct*, *alti*, and *far* under these perturbations indicates that their contribution is not driven solely by outliers or extreme conditions.

The conditioning-set construction was also evaluated by varying the maximum feature limit and observing how the selected feature set changed under different levels of model sparsity. Because the forward selection procedure is primarily governed by a fixed maximum number of features, changes in stopping behavior primarily reflect how early weaker dependencies are excluded under stricter model constraints. When stricter thresholds were applied, only a subset of variables, primarily *drct* and *far*, remained in the selected set. Under more relaxed thresholds, additional variables such as *sknt* and *purposeofflight* were included. This behavior reflects the tradeoff between model sparsity and sensitivity to weaker dependencies, and further supports the interpretation that the strongest relationships are concentrated among a small subset of features.

Bootstrap resampling results provide a complementary view of robustness by quantifying how frequently each feature is selected across repeated samples of the dataset. Table 15 summarizes the selection frequency of candidate variables across bootstrap iterations.

Feature	Selection Frequency
weathercondition	0.100
purposeofflight	0.800
far	0.920
drct	1.000
dwpf	0.280
sknt	0.380
operator	0.480
alti	1.000
tmpf	0.480
make	0.560

Table 15: Bootstrap Stability Candidate Features

The results show that *drct* and *alti* are selected in all 50 bootstrap samples, while *far* and *purposeofflight* also exhibit high selection frequencies. These variables can therefore be interpreted as robust precursors under the definition introduced in Section 7.4, as they remain both statistically significant under permutation testing and stable under resampling. In contrast, variables such as *dwpf* and *weathercondition* exhibit lower selection frequencies, indicating that their contribution is more sensitive to sampling variability and may reflect weaker or context-dependent relationships.

Taken together, these findings suggest that the most stable informational structure in the dataset is driven by a combination of environmental variables related to wind direction and atmospheric pressure, along with operational variables associated with regulatory classification and flight purpose. This combination provides partial support for Research Question 4, indicating that both environmental and operational factors contribute to distinguishing higher-risk runway excursion conditions. However, the absence of strong and consistent contributions from variables such as precipitation or visibility suggests that no single environmental factor dominates the predictive structure. Instead, runway excursion risk appears to arise from the interaction of multiple moderate-strength dependencies rather than a single dominant predictor.

The sensitivity analyses indicate that a core subset of variables exhibits consistent informational contribution across

multiple perturbations, while other features are more sensitive to modeling choices and sampling variation. These results support the interpretation of the identified precursors as stable but not exhaustive representations of the underlying dependency structure.

9 Discussion

The results of this analysis provide a structured view of how environmental and operational variables contribute informatively to runway excursion severity, but they must be interpreted with care. The inferred dependencies reflect conditional information flow under the modeling assumptions of mutual information, conditional mutual information, and causation entropy, rather than definitive physical causation. As such, the edges identified in the dependency graph represent statistical relationships supported by the available data and estimation procedures, but alternative explanations remain plausible.

Several of the inferred edges are consistent with domain knowledge regarding runway excursions. The strong conditional information contribution of *wind direction* (*drct*) aligns with established aviation safety findings indicating that crosswind conditions affect directional crosswind conditions affect directional control during landing and rollout, increasing the likelihood of veer-off events [6, 21]. Because crosswind and tailwind components were explicitly engineered from wind speed and direction, the importance of *drct* likely reflects these underlying aerodynamic effects in a more operationally meaningful form. Similarly, the prominence of *altimeter setting* (*alti*) may reflect broader atmospheric conditions, including pressure systems that are often associated with changing weather patterns such as storms or frontal activity, which can indirectly influence runway conditions and aircraft performance.

Operational variables also exhibit meaningful contributions. The importance of FAR classification (*far*) suggests that regulatory context and type of airport operation (e.g., Part 91 vs. Part 121) may capture differences in operational procedures, training, and safety margins. Similarly, *purpose of flight* (*purposeofflight*) may reflect distinctions between commercial, instructional, or personal operations, each of which may be associated with different risk profiles. These relationships are consistent with prior work highlighting the role of human and operational factors in runway excursion risk [19, 24].

At the same time, alternative explanations must be considered. Some variables may act as proxies for unobserved factors rather than distinct contributors. For example, *far* and *purposeofflight* may indirectly capture differences in pilot experience, aircraft weight, or operational constraints that are not explicitly included within the dataset. Similarly, environmental variables may be correlated with unmeasured runway conditions such as contamination (e.g., wet, icy) or braking effectiveness, which are known contributors to runway excursions but are not directly observed in the data [14]. These omitted variables introduce the possibility of confounding, where observed dependencies reflect underlying factors not captured in the analysis.

Measurement error and data limitations also affect interpretation. METAR observations provide point-in-time weather measurements that may not fully represent conditions at the exact moment of the runway excursion event, particularly given the temporal alignment challenges described in Section 10. Additionally, categorical variables with missing or generalized entries (e.g., "UNKNOWN") may introduce ambiguity and artificially influence feature selection. These factors may weaken or distort true relationships and should be considered when interpreting the results.

Causation entropy, while effective for identifying conditional dependencies, may still be misleading in certain contexts. In particular, unobserved variables, limited sample size, and potential nonstationarity in the data-generating process can all affect the reliability of inferred relationships [23, 12]. The greedy forward selection procedure may also favor variables that provide incremental information early in the selection process, potentially overlooking alternative combinations of predictors that jointly contribute to the outcome.

Additional data would help reduce ambiguity and strengthen interpretation. Incorporating direct measurements of runway surface conditions, aircraft performance characteristics (e.g., landing distance, braking action), and pilot-related variables (e.g., experience, training) would provide a more complete representation of the aviation system in regard to runway excursions. Higher-resolution temporal data would also improve the reliability of environmental features. Finally, access to larger datasets spanning multiple years and operational contexts would allow for more robust estimation of information-theoretic quantities and improved generalization of the results.

Overall, the findings should be interpreted as identifying candidate precursor relationships supported by the data under the modeling assumptions, rather than establishing definitive causal mechanisms. The results provide a structured

basis for hypothesis generation as well as further investigation into runway excursion risk.

10 Limitations

Several limitations of the dataset and methodology should be considered when interpreting the results of this analysis.

First, the dataset contains a relatively limited number of runway excursions with 428 observations, which constrains statistical power and may limit the detection of weaker underlying relationships. Information-theoretic estimators, particularly CMI, can exhibit high variance in small samples, which may lead to unstable estimates, missed dependencies (false negatives), or spurious associations (false positives).

Second, outcome variables and operational labels are imperfect. Variables such as *purpose of flight*, *weather condition*, and *probable cause* contain missing or generalized entries (e.g., "UNKNOWN"), which were retained as explicit categories. While this preserves sample size, it introduces ambiguity and may bias feature selection by artificially inflating the importance of missingness-related categories or masking true relationships.

Third, important confounding variables are not fully captured in the dataset. Factors such as runway condition (e.g., wet, icy), pilot experience, aircraft weight, approach stability, and braking performance are either missing or only indirectly represented. The absence of these variables may result in omitted variable bias, where observed relationships reflect unmeasured factors, potentially leading to incorrect causal interpretations or false positives in the inferred dependency structure.

Fourth, the temporal alignment between runway excursion events and METAR observations introduces uncertainty. Although the nearest-in-time weather report was selected, the mean time difference exceeds two days due to reporting irregularities, despite a median difference of only 10 minutes. This mismatch may weaken true environmental signals or introduce noise, potentially biasing estimates toward weaker or misleading relationships.

Fifth, the use of discretization for continuous variables introduces sensitivity to binning choices. While quantile-based binning helps preserve distributional structure, different bin sizes or discretization strategies may yield different MI/CMI estimates. This can affect feature rankings and the resulting causal structure, particularly for variables with skewed or heavy-tailed distributions.

Finally, class imbalance in the target variable (damage severity) may influence feature selection results. Because the majority of events in the dataset result in substantial damage, the model may prioritize features that distinguish common outcomes rather than rare but critical severe cases, potentially under-representing predictors of extreme events. Overall, these limitations suggest that results should be interpreted as exploratory and hypothesis-generating rather than definitive causal conclusions. Future work incorporating larger datasets, richer operational variables, and temporally precise measurements would help improve the robustness and interpretability of the findings.

11 Implementation and Reproducibility

The runway excursion analysis is implemented in Python using a modular pipeline designed to support both reproducibility and transparency [13]. The project is organized into a structured directory layout that separates data ingestion, preprocessing, feature engineering, exploratory analysis, and information-theoretic modeling. This organization allows each stage of the workflow to be independently executed and verified, and reused.

11.1 Codebase Structure

The full project structure is organized as follows:

- **data/**
 - **raw/**
 - * **NTSB Accident Data/**: NTSB Runway Excursion dataset
 - * **Weather Data/**: raw weather files, ASOS Stations list, and METAR fallback stations derived from ASOS using Haversine distance
 - **processed/**
 - * **Airport Stations/**: airport-station mappings and unmatched airport records

- * **METAR/**: combined and time-aligned METAR datasets matched to runway excursion events
- * **Merged Datasets/**: cleaned and final NTSB-METAR datasets used for analysis
- **pipeline/**: Contains preprocessing, exploratory data analysis (EDA), feature engineering, and visualization utilities. This includes generation of wind components (crosswind and tailwind) and distributional analysis of meteorological variables.
- **information_theory**: Custom implementations of entropy-based estimators, including mutual information (MI), conditional mutual information (CMI), and causation entropy (CSE). These form the core of the dependency estimation framework.
- **causal_analysis/**: Main causal discovery package that contains utilities to perform greedy forward selection using CSE and CMI, construction and visualization of directed dependency graphs, helper functions for target construction and feature preprocessing, and functions to convert results into publication-ready LaTeX tables.
- **METAR Scripts/**: scripts for METAR retrieval, station matching, and data merging with NTSB data
- **figures/**: saved visualization outputs in `.svg` format and `.png` format
- **Project Documentation/**: `.pdf` file of this report
- **Configuration files**: `requirements.txt` for required Python packages, `.env.example` for environment setup, `.gitignore` for version control, and `README.md`, which serves as the repository's documentation file.

The full codebase for this project, including data processing, feature engineering, and causal analysis pipelines, is publicly available on GitHub: <https://github.com/rachek253/Entropy-Based-Causal-Analysis-of-Precursors-to-Runway-Excursions>
`git`

11.2 Pipeline Execution

The full analysis pipeline performs the following steps:

1. Load raw NTSB accident data and METAR weather observations
2. Standardize airport identifiers and match events to nearest METAR station observations
3. Construct 24-hour METAR alignment windows and select temporally closest observations
4. Clean and preprocess dataset (remove redundant variables, handle missing values, standardize types)
5. Engineer derived meteorological features (crosswind and tailwind components)
6. Generate exploratory visualizations, summary statistics for numerical variables, as well as variable counts for categorical variables
7. Compute mutual information (MI) rankings using both custom and scikit-learn estimators
8. Perform conditional mutual information (CMI)-based greedy feature selection (terminated by `max_features` constraint and absence of positive causation entropy gain; not by statistical thresholds)
9. Execute causation entropy (CSE) forward selection for causal discovery
10. Run permutation testing for statistical significance of selected dependencies
11. Perform bootstrap resampling to evaluate feature stability
12. Construct directed causal graph from selected features
13. Export results as LaTeX tables and save figures for reporting

11.3 Reproducibility and Environment

The analysis is fully reproducible using a fixed random seed (`RANDOM_SEED = 0`) applied across NumPy and scikit-learn based resampling procedures. Environment configuration is handled through a `.env` file that specifies the root project directory (`PROJECT_DATA_DIRECTORY`), ensuring portability across systems.

All experiments were conducted in a Python environment using standard scientific computing libraries, including NumPy, pandas, scikit-learn, matplotlib, and NetworkX, along with custom information-theoretic modules that were developed for this project.

11.4 Computational Cost

The computational cost of the pipeline is moderate due to the use of an API to access METAR weather observations from the Iowa State Mesonet archive [9], bootstrap resampling [26], and permutation testing [18]. The most expensive components are the repeated estimation of mutual information-based quantities (MI, CMI, and CSE), which require recomputing empirical joint distributions across discretized feature spaces at each iteration of the greedy selection procedure. In addition, bootstrap resampling further increases computational load because the full causation entropy pipeline is executed repeatedly over many resampled datasets to evaluate stability. Permutation testing also contributes significantly to runtime, as each feature-target relationship must be re-evaluated under multiple random permutations to construct an empirical null distribution.

Despite these costs, the overall runtime remains manageable due to the relatively small sample size (428 events) and the use of vectorized NumPy operations and efficient pandas-based preprocessing. Memory usage is also controlled by avoiding storage of intermediate high-dimensional joint probability tables and instead recomputing distributions on demand during each iteration of the algorithm. As a result, the pipeline remains suitable for standard research computing environments without requiring distributed or high-performance computing resources.

12 Conclusion

This study aimed to identify candidate precursors to runway excursion severity using publicly available observation data and information-theoretic methods. By applying mutual information, conditional mutual information, and causation entropy within a structured feature selection and causal discovery framework, the analysis characterized conditional dependencies between environmental, operational, and aircraft-level variables and runway excursion outcomes.

The most defensible findings of this study are summarized as follows:

1. *Wind direction (drcet)* and *altimeter setting (alti)* consistently exhibit the strongest conditional information contribution to runway excursion severity.
2. Operational variables, particularly *FAR classification (far)* and *purpose of flight (purposeofflight)*, provide additional independent informational value beyond environmental factors.
3. A small subset of variables demonstrates high robustness across permutation testing and bootstrap resampling, indicating stable dependency structure under sampling variability.
4. Runway excursion risk appears to arise from interacting environmental and operational factors rather than a single dominant predictor, with moderate-strength dependencies distributed across multiple variables.

From a practical perspective, these results suggest potential applications in aviation safety monitoring and decision support. Variables identified as stable informational contributors, such as wind direction and operational classification, could be incorporated into risk assessment tools or early warning systems to highlight higher-risk conditions. For example, combining wind-related features with operational context may support improved situational awareness for pilots as well as air traffic controllers during critical phases of flight. Additionally, these findings may inform training programs and safety interventions by emphasizing the interaction between environmental conditions and operational decision-making.

Future work should focus on expanding both the dataset and methodological framework. Incorporating additional variables related to runway condition, aircraft performance, and pilot behavior would help reduce confounding and improve interpretability. Methodologically, exploring alternative estimators, incorporating temporal modeling, and validating findings across independent datasets would strengthen the robustness of the results. Further validation using predictive models or simulation-based approaches may also help assess the practical utility of the identified relationships.

This study provides an exploratory, data-driven characterization of informational dependencies associated with runway excursion severity. While not establishing causal mechanisms, the results identify a set of stable candidate precursors that may support further research and contribute to improved understanding of runway excursion risk in aviation safety contexts.

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